

WRIGHT AAF, GA, USA

WMO: 722090

Lat: 31.883N

Lon: 81.567W

Elev: 14

StdP: 101.16

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03875

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	-0.9	-11.3	1.4	3.3	-8.8	1.8	3.7	9.5	12.2	8.6	11.7	1.3	320	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	35.7	24.8	34.7	24.8	33.6	24.7	27.0	31.8	26.4	31.5	25.8	30.8	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.2	28.9	25.2	20.3	28.3	24.7	19.7	27.8	84.7	31.8	82.0	31.5	79.7	31.2	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.9	5.7	DB	-5.5	37.2	1.8	1.1	-6.8	38.0	-7.8	38.6	-8.9	39.2	-10.2	40.0	
			WB	-6.7	27.8	1.8	0.8	-8.0	28.4	-9.1	28.9	-10.1	29.3	-11.4	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	10.2	12.4	15.4	19.1	23.4	26.9	27.9	27.7	25.4	20.4	14.4	12.7	(6)
		DBStd	7.29	5.36	5.01	4.57	3.67	2.87	1.99	1.63	1.56	2.14	3.85	4.39	5.01	
		HDD10.0	146	66	30	9	1	0	0	0	0	0	11	30		
		HDD18.3	917	256	173	112	35	2	0	0	0	24	131	183		
		CDD10.0	3689	71	97	178	273	415	507	556	548	461	324	144	114	
		CDD18.3	1418	3	7	22	58	159	257	298	290	211	89	14	9	
		CDH23.3	13182	19	64	208	588	1471	2523	3013	2782	1710	671	88	44	
		CDH26.7	5577	1	6	36	160	593	1204	1449	1255	684	179	8	2	
(14)	Wind	WSAvg	2.2	2.3	2.5	2.6	2.6	2.4	2.2	2.1	1.9	1.9	2.0	2.0	2.0	(14)
(15)	Precipitation	PrecAvg	1264	87	88	95	71	82	165	141	159	117	105	52	72	(15)
		PrecMax	1610	280	231	244	178	201	263	251	325	295	353	138	195	
		PrecMin	867	11	19	9	11	21	50	53	76	36	5	2	16	
		PrecStd	221	60	48	54	43	49	60	47	47	59	89	39	44	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	25.6	27.6	29.5	31.7	36.1	36.6	36.9	36.7	35.0	32.0	27.9	26.6		(19)
		0.4% MCWB	19.1	19.6	18.6	20.7	21.9	24.1	25.3	25.7	23.7	23.9	21.0	21.4		
		2% DB	23.2	25.3	27.7	30.2	33.4	35.1	35.6	35.1	33.0	30.2	25.8	24.6		
		2% MCWB	17.6	18.4	18.9	20.3	22.1	24.7	25.2	25.7	24.1	22.8	19.5	19.8		
		5% DB	20.8	23.4	25.7	28.7	31.6	34.0	34.4	33.8	31.8	28.6	24.1	22.7		
		5% MCWB	16.4	17.2	18.0	19.9	21.8	24.6	25.1	25.5	24.0	21.8	18.8	19.1		
		10% DB	19.0	21.0	23.8	27.0	30.1	32.6	33.2	32.6	30.6	27.3	22.3	20.7		
		10% MCWB	15.8	15.9	16.9	19.1	21.6	24.5	25.0	25.4	23.8	21.1	17.9	18.3		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	20.6	21.0	21.0	23.6	25.2	27.4	27.4	28.3	26.9	25.5	23.2	22.9		(27)
		0.4% MCDB	22.9	24.7	25.9	28.2	29.7	32.4	32.1	32.8	30.7	28.7	25.5	25.3		
		2% WB	19.3	19.7	20.1	22.5	24.1	26.5	26.7	27.3	25.6	24.5	21.8	21.3		
		2% MCDB	21.6	23.7	24.8	26.8	28.5	32.1	32.1	31.7	29.7	27.5	23.5	22.9		
		5% WB	18.1	18.5	19.2	21.5	23.4	25.8	26.1	26.6	25.1	23.7	20.5	20.0		
		5% MCDB	20.1	22.3	23.6	25.7	28.2	31.3	31.7	31.2	29.4	26.6	22.4	21.5		
		10% WB	16.4	17.2	18.3	20.5	22.8	25.2	25.6	26.0	24.6	22.9	18.9	18.6		
		10% MCDB	18.6	19.9	22.5	24.7	27.6	30.5	31.1	30.5	28.5	25.7	21.8	21.0		
(35)	Mean Daily Temperature Range	MDBR	12.1	12.8	13.2	13.5	12.3	11.0	10.5	9.8	10.0	11.8	12.8	11.7		(35)
		5% DB MCDBR	13.8	14.4	14.7	14.9	14.3	12.6	11.9	11.4	11.8	12.3	13.4	12.1		
		5% DB MCWBR	8.7	8.3	6.9	6.6	5.1	4.3	3.8	3.8	4.1	5.9	7.8	7.9		
		5% WB MCDBR	11.3	12.3	12.7	12.7	11.5	11.4	10.6	10.2	9.4	9.6	10.2	10.8		
		5% WB MCWBR	8.5	8.1	6.8	6.5	4.8	4.5	4.0	4.1	3.7	5.2	7.3	8.2		
(40)	Clear-Sky Solar Irradiance	taub	0.331	0.339	0.374	0.401	0.446	0.496	0.522	0.512	0.428	0.394	0.355	0.343		(40)
		taud	2.513	2.491	2.416	2.334	2.241	2.156	2.094	2.124	2.358	2.416	2.479	2.509		
		Ebn at Noon	893	920	907	890	848	803	780	783	842	847	856	859		
		Edh at Noon	87	97	112	127	140	152	161	155	118	104	89	83		
(44)	All-Sky Solar Radiation	RadAvg	2.99	3.59	4.73	5.90	6.36	6.22	6.17	5.58	4.75	4.16	3.30	2.62		(44)
		RadStd	0.16	0.37	0.43	0.39	0.37	0.48	0.39	0.35	0.39	0.43	0.22	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.56	+0.78	N/A	N/A	N/A	N/A	N/A	-84	+191	+133

Nomenclature: See separate page